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Predicting the compressive strength of concrete with fly ash admixture using machine learning algorithms

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ABSTRACT

The cementitious composites have different properties in the changing environment. Thus, knowing their mechanical properties is very important for safety reasons. The most important in the case of concrete is the Compressive strength (CS). To predict the CS of concrete Machine learning (ML) approaches has been essential. This study includes the collection of data from the experimental work and the application of ML techniques to predict the CS of concrete containing fly ash. The chemical and physical properties of all the materials used in this study were evaluated. Although, the emphasis of this research is on the use of supervised machine learning algorithms to forecast the CS of concrete. The Gene expression programming (GEP), Artificial neural network (ANN), and Decision tree (DT) algorithms were investigated for the prediction of outcome (CS). Concrete samples (cylinders) with different mix ratios were cast and tested at various ages to maintain the required data for applying it to run the models. Total 98 data points were collected from the experimental approach, in which seven parameters namely cement, fly ash, superplasticizer, coarse aggregate, fine aggregate, water, and days were taken as input to predict the output which was CS parameter. The experimental data is further validated by mean of k-fold cross-validation using R^2, root mean error (RME), and Root mean square error (RMSE). In addition, statistical checks were incorporated to evaluate the model performance. In comparison, the bagging algorithm shows high accuracy towards the prediction of outcome as indicated by its high coefficient correlation (R^2) value equals to 0.95, while R^2 value for GEP, ANN and DT comes to 0.86, 0.81 and 0.75 respectively.

1. Introduction

Cement industries are one of the biggest carbon dioxide (CO_2) emission producers around the world. Hence have a magnificent effect on the environmental conditions [1]. The production of greenhouse gases (GHG) is mainly due to the production and utilization of cement in construction projects every year all around the world [2]. Moreover, cement production is around four billion tons yearly and about one ton of cement emits the same amount of CO_2 gases [3]. Production of Portland cement (PC) results in seven percent of all CO_2 emissions to the atmosphere. The Calcination of calcium oxide (CaO) plays a vital role in the production of CO_2 during the manufacturing of PC [4,5]. To minimize this effect use of waste or recycled material in concrete production is recommended [5]. It will be not only suitable for the requirement of concrete but also minimalize the degradation of the natural environment [6]. PC can be replaced by various supplementary materials obtained as wastes from the industrial processes (e.g. ground granulated blast furnace slag (GGBS) [7], granite powder [8], and fly ash [9]). Using these supplementary raw materials will not only improve the properties of hardened concrete but also reduce the carbon footprint. It has been observed that the use of GGBS, granite powder and fly ash in the replacement of PC within concrete can minimize 80% of GHG emission [10]. Such waste materials are normally stored in heaps, which also affects the environment. The application of these waste materials as a partial replacement of cement is also beneficial from this point in the

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Notations

AI	Artificial Intelligence
ANN	Artificial neural network
GEP	Genetic engineering programming
HPC	High performance concrete
MSE	Mean square error
DL	Deep learning
FA	Fly ash
SVM	Support vector machine
GB	Gradient boosting
RHA	Rice husk ash
SF	Silica fume
RF	Random forest
ML	Machine learning
GGBS	Ground granulated blast furnace slag
DT	Decision tree
MAE	Mean absolute error
RMSE	Root mean square error
SCC	Self-compacting concrete
POFA	Palm oil fuel ash
IREMSVM-FR	Intelligent rule-based enhanced multiclass support vector machine
ANFIS	Adaptive Neuro Fuzzy Inference System
RMS	Response surface method
RKSA	Random Kitchen Sink Algorithm
MOGW	multi-objective grey wolves

construction industry [11].

The importance of concrete material can be judged by its usage all over the world every year, as it is the second-most utilizing material after water on the earth [12]. Concrete is used as a building material all over the globe because of its benefits by various properties like strength, durability, stiffness, hardness, porosity, density, fire resistance, and thermal resistance. Among these, the compressive strength is thought to be the most important as it strongly affects the concrete element's safety and durability [13]. The reason for the different values of the compressive strength of concrete is the fact, that concrete is a heterogeneous material made up of a binder, aggregate, water, and admixtures [14]. All these ingredients and their mixtures affect the compressive loading capacity of concrete, including water to binder ratio, size of aggregate, type of binder, or composition of waste [15]. It is difficult to find or predict the compressive strength of concrete accurately in such a complicated mixture. The compressive strength of concrete can be evaluated in the laboratory during the tests by crushing the cylinders or cubes of standard dimension for a required time after the casting of the sample [16]. This method is standardized all over the world. However, the laboratory tests are ought to be nowadays inefficient and uneconomical as it is costly and time-consuming. Recently, with the evolution of Artificial intelligence (AI), Machine learning algorithms (ML) are used to predict many mechanical properties in concrete [17]. ML techniques like regression, clustering, and classification can be used to estimate the various other parameters with different efficiency and help in forecasting the compressive strength with accurate precision [18-23].

2. Machine learning overview

Artificial intelligence (AI) is further classified into ML with the supreme focus on the development of predictive algorithms. This is due to the unbiased identification of various patterns having large datasets without being a direct arrangement for a given task. This area of artificial intelligence allows computers to execute those complicated and complex tasks which were nuanced for the machines. These algorithms came up with the ability to learn the patterns from the data rather than by straightforward programming by humans. The idea of these algorithms is generally based on the training, which enables the computer to learn the properties/features that preferably constitute the data set for the given problem and then interpreting this information to produce explanations of more available datasets. For example, if the selected algorithm is trained to transform a benign from a spiteful lesion on imaging and can be applied to other image data and categorize lesions as either benign or malignant, depending on previously "learned" criteria. Fig. 1 shows the hierarchy and subfields of artificial intelligence. Based on the piece of work, ML models can be broadly categorized as follow:

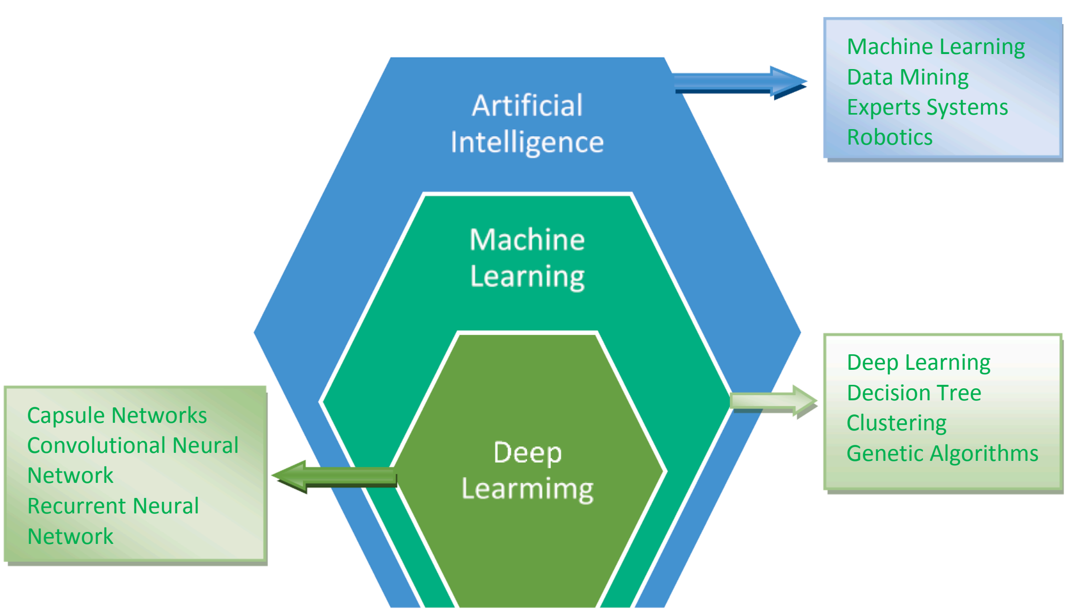

Fig. 1. Indicating the hierarchy and subfields of the artificial intelligence.

2.1. A. Supervised learning

In the supervised learning-based tasks, the presentation of the model is with the labeled dataset, which is also called a feature vector, which means that the datasets having examples of observations along with their expected outcomes. These models are enabled to produce an inferred function which maps the features vectors for the output labels. The commonly used and popular supervised machine learning techniques are decision tree, support vector machine, AdaBoost, bagging, boosting, artificial neural network, gene expression programming.

2.2. B. Unsupervised learning

In the case of unsupervised learning, the given datasets have very limited or sometimes with zero levels of information for the output labels. Instead, the aim of these models is to obtain the relationship among the observations and/or reveal the inactive parameters. Some examples of unsupervised learning-based approaches are k-means, Self-Organizing Maps, and Generalized Adversarial Networks (GANs).

3. Machine learning techniques with literature review

Machine learning techniques such as Artificial neural networks (ANN), Decision trees (DT), Support vector machines (SVM), Genetic engineering programming (GEP), and Deep learning (DL) are now widely used to predict different scientific problems [24]. To overcome the shortening of these techniques the design of SVM is such a way that can deal in a better way with the nonlinear regression problems [25]. It has a well-generalized ability and can achieve a better global optimal solution instead of a local optimum. While the Decision tree (DT) and Random forest (RF) behaves like tree structure uses roots and nodes for prediction the results [26]. DT uses a complete database with the variable of its interest whereas the selection of RF is random for features variables within the parameters that generate the number of trees in prediction. Prediction is then averaged and connected with maximum voters shows an accurate prediction. GEP is one of the latest ML computer-based programs via mimicking the Darwinian evolution process [27]. It expresses the non-linear relationship as an expression type tree. Machine learning (ML) approaches are normally used to extract unknown patterns, data information, and relationship from a large database. Although, this process uses the technology of database along with machine learning as well as statistical analysis. There are two techniques used for modeling and prediction. One of them is a standard approach situated on a single separate model, and another is known as the ensemble algorithm approach [28]. As per the previous research on these algorithms, ensemble techniques presents to be more accurate in

Table 1
The ML algorithms for different mechanical properties prediction.

Sr. No	Algorithm name	Notation	Dataset	Prediction properties	Year	Waste material used	References
1.	Individual and ensemble algorithm	GEP, DT and Bagging	270	Compressive Strength	2021	FA	[21]
2.	Individual with ensemble modeling	ANN, bagging and boosting	1030	Compressive strength	2021	FA	[22]
3.	Individual Algorithms	ANN, GEP, DT	642	Chloride Concentration	2021	FA	[18]
4.	Data Envelopment Analysis	DEA	114	Compressive strength Slump test L-box test V-funnel test	2021	FA	[42]
5.	Multivariate	MV	21	Compressive strength	2020	Crumb rubber with SF	[43]
6.	Support vector machine	SVM	–	Compressive strength	2020	FA	[44]
7.	Support vector machine	SVM	115	Slump test L-box test V-funnel test Compressive strength	2020	FA	[45]
8.	Adaptive neuro fuzzy inference system	ANFIS with ANN	7	Compressive strength	2020	POFA	[46]
9.	Gene expression programming	GEP	277	Axial capacity	2020	–	[20]
10.	Gene expression programming	GEP	357	Compressive strength	2020	–	[47]
11.	Random forest and gene expression programming	RF and GEP	357	Compressive strength	2020	–	[48]
12.	Artificial neuron network	ANN	205	Compressive strength	2019	FA GGBFS SF RHA	[49]
13.	Intelligent rule-based enhanced multiclass support vector machine and fuzzy rules	IREMSVM-FR with RSM	114	Compressive strength	2019	FA	[50]
14.	Random forest	RF	131	Compressive strength	2019	FA GGBFS FA	[51]
15.	Multivariate adaptive regression spline	M5 MARS	114	Compressive strength Slump test L-box test	2018	FA	[52]
16.	Random Kitchen Sink Algorithm	RKSA	40	V-funnel test J-ring test Slump test Compressive strength	2018	FA	[53]
17.	Adaptive neuro fuzzy inference system	ANFIS	55	Compressive strength	2018	–	[54]
18.	Artificial neuron network	ANN	114	Compressive strength	2017	FA	[55]
19.	Artificial neuron network	ANN	69	Compressive strength	2017	FA	[56]
20.	Artificial neuron network	ANN	169	Compressive strength	2016	FA GGBFS FA RHA	[57]
21.	Artificial neuron network	ANN	80	Compressive strength	2011	FA	[58]
22.	Artificial neuron network	ANN	300	Compressive strength	2009	FA	[59]

comparison with the standard individual ML models [29]. Ensemble learning models use to train so-called weak learners initially using training data, then integrate the weak learner into a strong one [30].

Machine learning algorithms are used to predict the performance of various parameters for so many years. However, the trend of their use in civil engineering increases significantly in the last few years. It is due to their high accuracy level of mechanical properties prediction as presented in Table 1. ML working principle is similar to the conventional algorithms but because nonlinear behavior is more accurate as compared to linear behavior. Artificial neural networks (ANN), Decision trees (DT), Support vector machines (SVM), Random forest (RF), Gene expression programming (GEP), Deep learning (DL) are commonly used for the analysis of concrete mechanical properties [31]. Jesika et al. [32] used eleven algorithms for the shear strength prediction of the steel fibers reinforced concrete beams. Behnood et al. [33] used ANN with optimizer as multi-objective grey wolves (MOGW) to predict the mechanical properties of the silica fume concrete. Kadir et at. [34] used DT, ANN, SVR, and LR for the prediction of concrete compressive strength. Getahun et al. [35] used ANN algorithm to predict the compressive strength and tensile strength of waste concrete. Ling et al.[36] predict the compressive strength of concrete in marine nature using SVM and compared the result with ANN and DT models. Zaher et al. [37] predict the compressive strength of lightweight foamed concrete with various machine learning approaches. Taffese et al. [38] also used a machine learning algorithm for the assessment of the durability of reinforced concrete structures. Suguru et al. [39] made the development of automatic detector of cracks in concrete structures with machine learning. Photographs of the concrete samples were used for learning data, while for the crack detection deep learning was applied. Wassim et al.[40] evaluate the accuracy level of the machine learning models. In addition, Miao et al.[41] uses MLR, SVM, and ANN to predict the interfacial bond strength between fiber-reinforced polymers (FRPs) and concrete.

4. Research significance

The significance and novelty of this research are (1) conducting experimental work for fly ash-based concrete (FAC) through ASTM standards, and (2) making FAC model using the Machine learning (ML) algorithms. Moreover, the focus of this study is based on the prediction and comparison of concrete (fly ash based) compressive strength via supervised machine learning approaches. The DT, ANN, and boosting ML approaches were employed to predict and compare the outcome with the actual results. The boosting approach was used for optimization by developing twenty sub-models to have a more accurate value of the coefficient of correlation (R^2). These applications were used to compare the prediction performance of each technique. The importance of this research is to understand the role of input parameters and the accuracy level of the outcomes from the various ML algorithms. This study also presents the comparison approach of both ensemble and individual ML approaches with the results obtained for the experimental work. Each model performance was also evaluated from the statistical checks and k-fold cross-validation.

5. Description of data with the experimental procedure

The various properties of the materials used in the preparation of 98 mixes such as cement, fly ash, fine aggregate and coarse aggregate has been conformed to the specification as per the ASTM standards as shown in Tables below. Ordinary Portland cement (Type-1) was used in this experimental work. The information stated in ASTM C150 for cement has been adopted during the entire investigation of this research. The airtight polythene sheet was used to cover the bags of cement and prevention was made from the effect of moisture. The physical and chemical properties of cement and fly ash used in the research are listed in Table 2 and Table 3, respectively. The fine aggregate was tested as per the ASTM standard to determine its properties. Locally available coarse aggregates with a maximum nominal size of 25.4 mm have been used for preparing the concrete mix according to ASTM standards. The sieve analysis and physical properties of the coarse aggregate (CA) were evaluated according to ASTM recommendations and the results are listed in Table 4. In addition, the physical properties of fine and coarse aggregate can be seen in Table 5. Super-plasticizer was added to the concrete mix by weight of cement to achieve the desired flow. The fly ash used was obtained from Pakistan.

Table 2
Physical properties of Ordinary Portland Cement.

Physical Property	Content
Insoluble residue (% mass)	0.49
Specific surface area (cm^2 /g)	8300
Specific gravity (g/cm^3)	3.15
Particle size (d_{50}) (cm)	1.649
Loss on ignition (% mass)	2.21

Table 3
Chemical properties of Cement and Fly Ash.

Compound details	Cement	Fly ash
Silica (SiO_2)	18.97	52.1
Alumina (Al_2O_3)	6.96	4.26
Calcium Oxide (CaO)	65.81	2.32
Iron Oxide (Fe_2O_3)	3.62	26.9
Sodium Oxide (Na_2O)	0.11	0.110
Magnesium Oxide (MgO)	1.96	1.51
Potassium Oxide (K_2O)	0.46	1.48
Loss on Ignition		1

5.1. Test program

The details of mix proportion in terms of weight for all studied formulations are mentioned in Annexure A. A total of 98 cylindrical specimens of 200 mm height and 100 mm diameter were prepared. A w/c ratio ranging from 0.40 to 0.60 is adopted by the hit and trial method and varying superplasticizer demand third generation to achieve desire workability. Specimens were de-molded after 24 h and water cured at 23 °C for 28 days. Compressive strength test was performed as per ASTM C39 at 3, 7, 14, and 28 and 90 days to evaluate the strength progression of designed formulations to achieve a desire strength of 28 MPa [60]. The reflection of the experimental work can be seen in Fig. 2.

5.2. Modeling aspect description

The data includes in the supplementary file as illustrated in Annexure A. The dataset consists of 7 inputs (cement, fly ash, water, superplasticizers, coarse aggregate, fine aggregate, and days) with one output (concrete compressive strength). The relative frequency distribution of all the parameters is presented in Fig. 2. The statistical description of the parameters used during numerical analyses is listed in Table 6 and Table 7, respectively.

Table 4
Physical properties of Fly ash.

Property	Value range
Retention on 45 µm Sieve, %	< 34
Lime Reactivity, N/mm^2	6
Soundness by Autoclave Expansion, %	0.05
Drying Shrinkage, %	0.06
Compressive Strength (as percent of strength of corresponding plain cement mortar cubes), %	80

Table 5
Physical properties of Fine Aggregate & Coarse Aggregate.

Property	Fine Aggregate	Coarse Aggregate	Standards Followed
Bulk Specific Gravity	2.63	2.70	ASTM C128/ C127
Moisture Content, %	1.129	0.65	ASTM C566
Moisture Absorption, %	1.06	1.36	ASTM C128/ C127
Fineness Modulus	2.41	–	ASTM C136
Nominal Maximum Size, mm	–	25.4	–
Rodded Unit Weight, kg/m^3	–	1550.6	ASTM C29

6. Methodology in making models

This section describes the algorithms used in modeling to predict the compressive strength of fly ash-based concrete. The anaconda software was used to run the models [61]. The forecasting of the compressive strength is made by using ensemble algorithm (boosting) and individual approaches like Artificial neuron network (ANN), with a tree-based algorithm such as Decision tree (DT) and Genetic engineering programming (GEP). The detailed flowchart of the algorithm used is depicted in Fig. 3.

The decision tree is considered one of the simplest and strong algorithms for classified data. It is a tree-like model which separates data points into numerous classes depend on the data points that meet certain criteria. DT uses the properties of input data as criteria to manage the classification task. The decision tree behaves in such a manner that the criteria have zero similarity for both classification and regression trees. The schematic diagram of DT can be seen in Fig. 4.

On the other hand, artificial neural networks (ANNs) can learn like the brain of a human being. ANN is the base of Artificial intelligence (AI) which can solve complex problems easily. The schematic functionality of the ANN is described in Fig. 5. Using self-learning capabilities, ANN can generate better results in the form of output by performing learning processes. As human also requires standard rules or guidelines to come up with the appropriate result or output, ANNs also use a standard set of learning rules known as backpropagation. The parameter used in making an ANN model is listed in Table 8.

Gene expression programming (GEP) was initially presented in the form of genetic programming by Ferreira [62]. GEP can collect more data rapidly than the older genetic algorithms. It gives more transparency because genetic operators are working at the chromosome level. The basic steps/flowchart of GEP are graphically represented in Fig. 6. The process starts using the arbitrarily generated chromosomes of the first population. The chromosomes are then revealing, and the strength of every individual is set based on fitness cases. These individuals are then chosen as per their fitness for reproduction and modification. The new chromosomes are subjected to the same processes of development: expression of the genomes, conflict of the selection environment, selection, and then reproduction with changes till satisfactory results have been found [63].

The boosting regressor was initially invented by the estimation of the learning theories and then was reanalyzed by the machine learning researchers. It is considered an ensemble method (weighted average data of the predictions of individual classifiers). The basic boosting ML approach needs the explanation of two parameters. The number of splits (number of nodes) can fit each regression during the process. The total number of nodes equals the number of splits plus one. Identifying one split corresponds to an additional model remaining only with main effects. Identifying two splits corresponds to the main effects of the model and two-way interconnections and vice versa. Boosting adopts the same procedure as other ensemble ML algorithms. There is the splitting of data into a training dataset and a test dataset. The selected percentage of data is normally used as training data and the remaining percentage is used as the test data. The selection of percentages can be changed to obtain better accuracy.

7. Results and discussion

7.1. Decision tree (DT) model outcome

The results of the prediction result of FA-based concrete via decision

Table 6
Range of input and output variables.

Parameters	Abbreviation	Units	Minimum value	Maximum value
Cement	C	kg/m^3	136.1	376
Fly ash	FA	kg/m^3	92.1	168.3
Super plasticizer	SP	kg/m^3	0	18
Water	W	kg/m^3	141.1	220.5
Fine aggregate	FA	kg/m^3	687	905.4
Age	A	Days	3	90
Coarse aggregate	CA	kg/m^3	801	1118
Compressive Strength	CS	MPa	9.49	72.11

Fig. 2. Part of the specimens from the experimental work predicted to compressive tests.

Table 7
Descriptive analysis of parameters.

	Cement	Flash	Water	Superplasticizer	Coarse aggregate	Fine aggregate	Days
Mean	241.36	123.88	178.69	6.32	1002.24	794.55	33.79
Standard Error	5.64	1.03	1.84	0.5	7.24	4.67	2.94
Median	230.35	124.8	184.2	6.7	1014.85	794.9	28
Mode	213.5	124.8	191.3	0	1055.6	800	28
Standard Deviation	55.85	10.23	18.18	4.94	71.7	46.21	29.1
Sample Variance	3119.16	104.64	330.57	24.37	5140.86	2135.06	846.64
Kurtosis	−0.8	9.54	−0.61	−1.06	0.31	0.31	−0.19
Skewness	0.27	0.53	−0.07	0.02	−0.81	0.42	1.04
Range	239.9	76.2	79.4	18	317	218.4	87
Minimum	136.1	92.1	141.1	0	801	687	3
Maximum	376	168.3	220.5	18	1118	905.4	90
Sum	23653.5	12140.1	17511.6	619.3	98219.3	77866.3	3311
Count	98	98	98	98	98	98	98

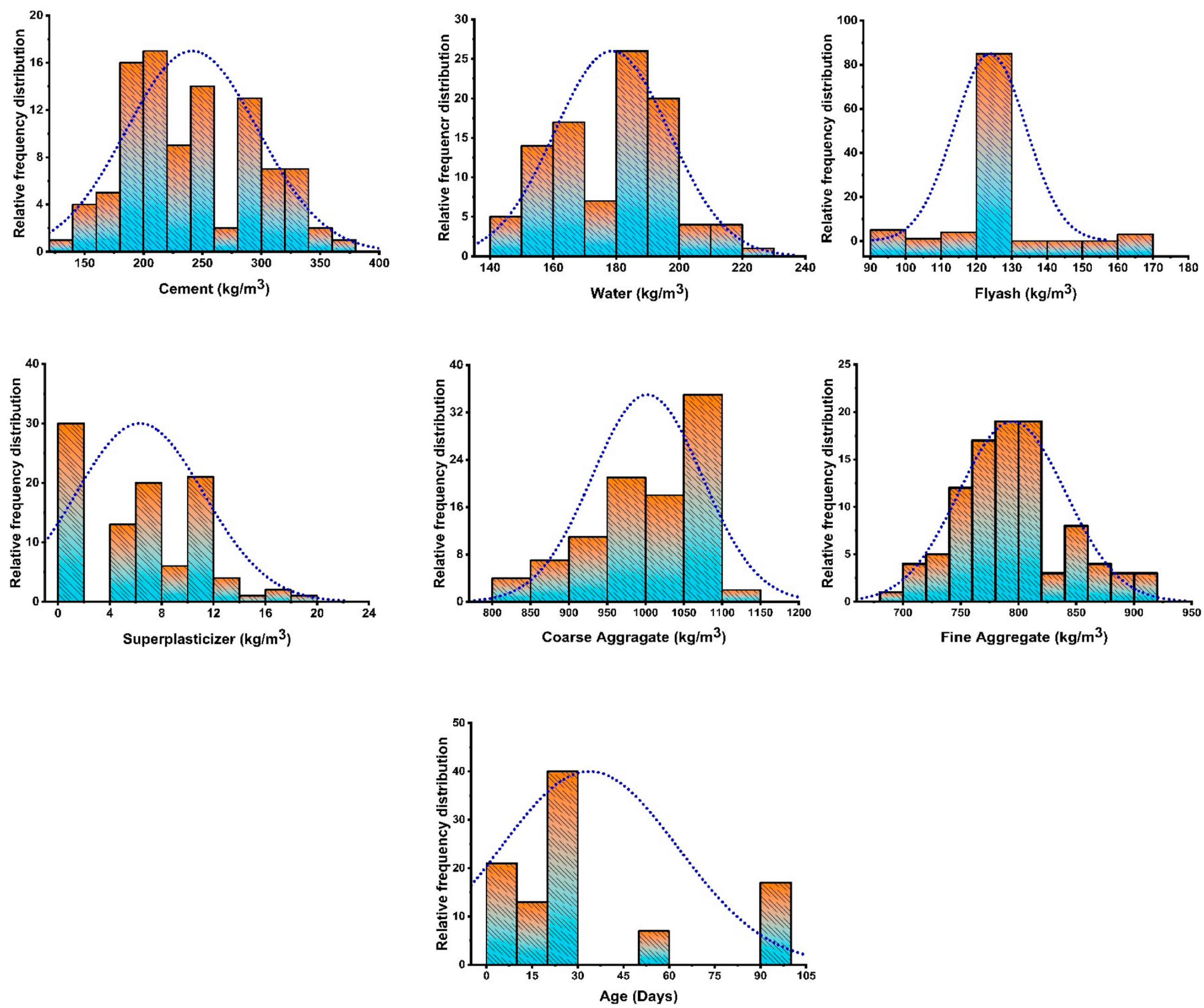

Fig. 3. Histogram showing the maximum concentration used in evaluating the compressive strength.

tree have been presented in Fig. 7. It can be seen that the DT accurately forecasts the concrete compressive strength with a better coefficient of determination $R^2 = 0.75$ as shown in Fig. 7(a). Moreover, the model error distribution illustrates that the average error of the test set is 4.21 MPa with the minimum and maximum error values of 0.002 MPa and 21.39 MPa respectively as depicted in Fig. 7(b). Also, the model accuracy can be judged by its output results as depicted in Table 9. It shows that most of the data points lie below 5 MPa with 65 percent accuracy and 30 percent of data lies in the range of 5 MPa to 10 MPa and one data point shows out space with 21.39 MPa.

7.2. *Artificial neural network (ANN) model outcome*

The prediction of the compressive strength by the ANN algorithm yields an adamantly strong relationship between the predicted and experimental value as shown in Fig. 8(a). Model shows an R^2 value of

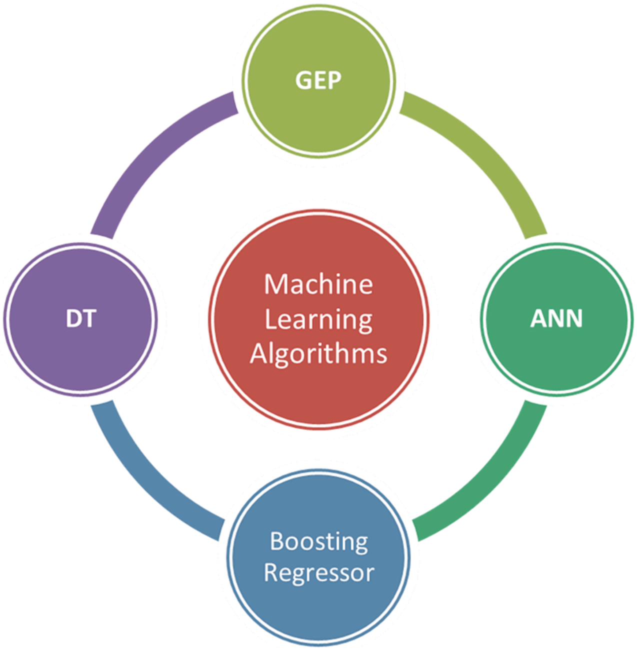

Fig. 4. Algorithm Flowchart.

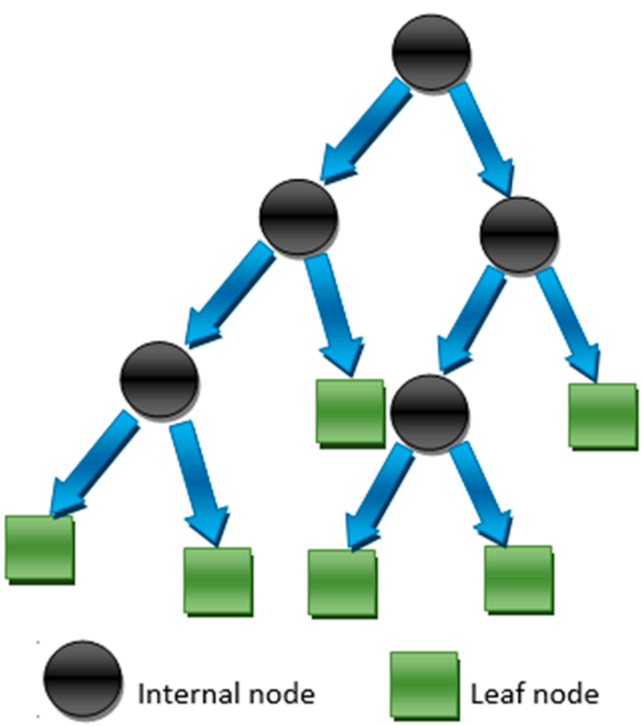

Fig. 5. The schematic diagram of a decision tree.

0.81 and a low error distribution presented in Fig. 8(b) with the average error of the training set was 2.96 MPa, while the minimum and maximum error were observed to be 0.01 MPa and 21.01 MPa. Similarly, 90 percent of the result of distribution error of model lies below 10 MPa as depicted in Fig. 8(b) and Table 9.

7.3. Genetic engineering programming (GEP) model outcome

The influence of variables in the prediction of compressive strength through linear regression is illustrated in Fig. 9. The GEP performed more accurately than the previous algorithms by showing a higher value of the coefficient of determination with R^2 value equals 0.86 as shown in Fig. 9(a). In addition, Fig. 9(b) represents the error distribution, and the average value of errors equals 3.35 MPa, with the minimum error value equal to 0.76 MPa, and the maximum error value equal to 5.98 MPa. Moreover, the predicted error result outcome shows that 100 percent of data points lie below 10 MPa which depicts the model accuracy with maximum predicted strength.

Table 8
ANN model parameters.

Parameters	Values
Hidden layer sizes	20,20
Activation function	relu
solver	Adam
alpha	0.0001
batch size	32
learning rate	Constant
Learning rate initial	0.001
Max iteration	10,000

7.4. Boosting regressor outcome

The prediction of the compressive strength of concrete via ensemble (boosting) ML algorithm shows much better performance as compared to individual ML approaches employed in this study. The reflection of its performance can be seen in Fig. 10(a) showing the correlation between the actual and targeted outcome. While Fig. 10(b) gives the information of its error distribution indicating the maximum and minimum values of the error equal to 3.03 MPa and 0.57, respectively with average error of about 2.013 MPa. Moreover, 100 percent of the error data lie below 5 MPa showing the accuracy of the model.

7.5. K-fold Cross validation and statistical metrics

K-fold cross validation algorithm represents the process of Jack's knife test which can be apply for various aspects, including for the reduction of biases in a random sampling of the training set, for holding out the data set, and to minimize the overfitting complications. Stratified ten-fold validation process is considered as precise and normally adopted for the optimum computational time. This study also approaches the same ten-fold using the splitting of data into k-group. It divides the data into 10 subsets, from which it uses nine out of ten subsets. The only one subset refer to validate the model. This process needs to repeat for ten times for the average outcome. In addition, the response of the applied models was also evaluated with the statistical checks. The following equations in accordance with the literature [64] were used to give the information of the model's performance (1–5).

$$RMSE = \sqrt{\frac{\sum_{i=1}^{n}(ex_i - mo_i)^2}{n}} \tag{1}$$

$$MAE = \frac{\sum_{i=1}^{n}|ex_i - mo_i|}{n} \tag{2}$$

$$RSE = \frac{\sum_{i=1}^{n}(mo_i - ex_i)^2}{\sum_{i=1}^{n}(\overline{ex} - ex_i)^2} \tag{3}$$

$$RRMSE = \frac{1}{e}\sqrt{\frac{\sum_{i=1}^{n}(ex_i - mo_i)^2}{n}} \tag{4}$$

$$R = \frac{\sum_{i=1}^{n}\left(ex_i - \overline{ex}_i\right)\left(mo_i - \overline{mo}_i\right)}{\sqrt{\sum_{i=1}^{n}\left(ex_i - \overline{ex}_i\right)^2 \sum_{i=1}^{n}\left(mo_i - \overline{mo}_i\right)^2}} \tag{5}$$

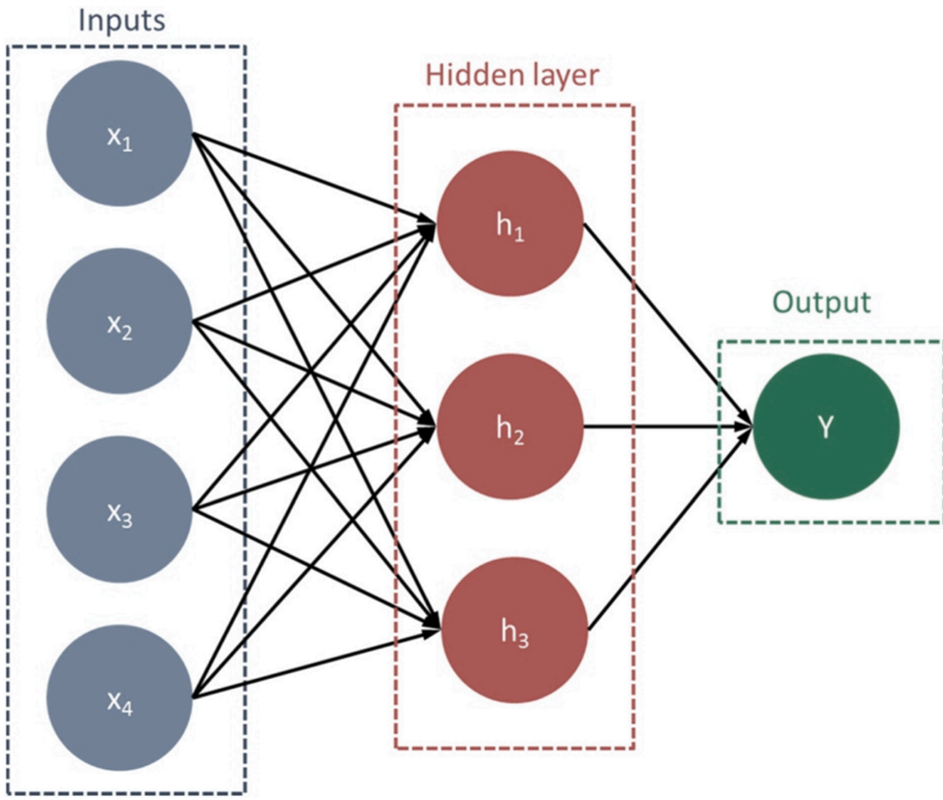

Fig. 6. Function of ANN.

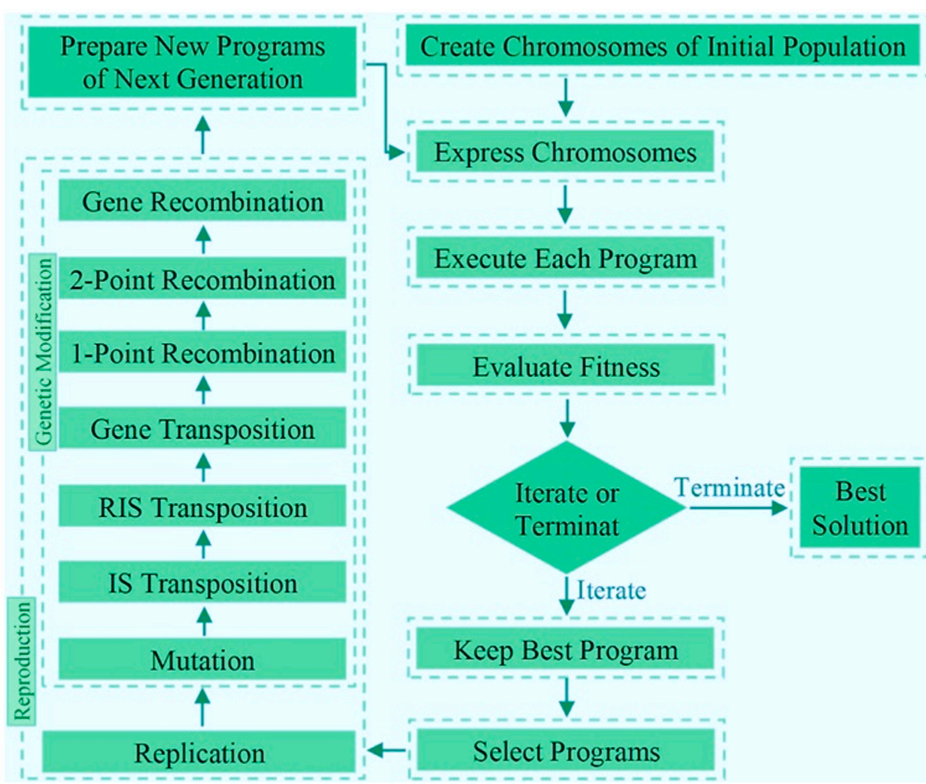

Fig. 7. The flowchart of gene expression programming.

Where,

ex_i = experimental value,

mo_i = predicted value,

$\overline{ex}_i$ = mean experimental value,

$\overline{mo}_i$ = mean predicted value obtained by the model,

Table 9
The predicted result of models with errors.

Data points	DT Y-Test	Y-Pred	Errors	ANN Y-Test	Y-Pred	Errors	GEP Y-Test	Y-Pred	Errors	Boosting Y-Test	Y-Pred	Errors
1	21.13	15.31	5.82	24.70	24.02	0.68	36.98	31.34	5.64	36.09	37.21	1.12
2	19.95	19.95	0.00	59.64	38.24	21.40	20.01	21.93	1.92	25.81	23.48	2.33
3	44.17	38.24	5.93	45.99	38.24	7.75	9.49	10.26	0.77	35.92	36.87	0.95
4	62.55	54.28	8.27	45.20	38.24	6.96	11.78	13.27	1.49	27.54	29.14	1.60
5	30.92	24.02	6.90	26.29	25.77	0.52	26.11	30.43	4.32	23.18	21.00	2.18
6	25.81	22.45	3.36	25.11	24.02	1.09	18.98	24.03	5.05	18.98	21.51	2.53
7	21.43	20.50	0.93	40.94	36.77	4.17	22.11	20.31	1.80	18.66	21.67	3.01
8	36.57	34.77	1.80	21.13	15.31	5.82	26.29	22.02	4.27	36.99	36.42	0.57
9	21.20	18.33	2.86	44.17	38.24	5.93	40.10	35.39	4.71	15.21	16.93	1.72
10	45.20	38.24	6.96	21.43	20.50	0.93	41.64	40.24	1.40	15.44	13.51	1.93
11	25.11	24.02	1.09	62.55	54.28	8.27	33.55	32.16	1.39	35.61	38.64	3.03
12	29.22	28.76	0.46	25.81	22.45	3.36	28.79	25.62	3.17	18.90	21.00	2.10
13	24.70	24.02	0.68	36.57	34.77	1.80	47.83	49.41	1.58	18.24	20.71	2.47
14	40.94	36.77	4.17	24.69	19.95	4.74	25.81	31.73	5.92	28.66	30.21	1.55
15	24.69	19.95	4.74	40.12	40.11	0.01	37.77	34.96	2.81	20.01	22.59	2.58
16	26.29	25.77	0.52	29.22	28.76	0.46	36.99	32.13	4.86	9.55	12.06	2.51
17	40.12	40.11	0.01	19.95	19.95	0.00	27.41	25.49	1.92	29.33	32.24	2.91
18	59.64	38.24	21.40	21.20	18.33	2.86	30.92	24.93	5.99	13.11	11.27	1.84
19	45.99	38.24	7.75	23.18	22.50	0.68	35.92	33.13	2.79	40.94	38.64	2.30
20	23.18	22.50	0.68	30.92	24.02	6.90	18.24	23.50	5.26	41.64	40.61	1.03

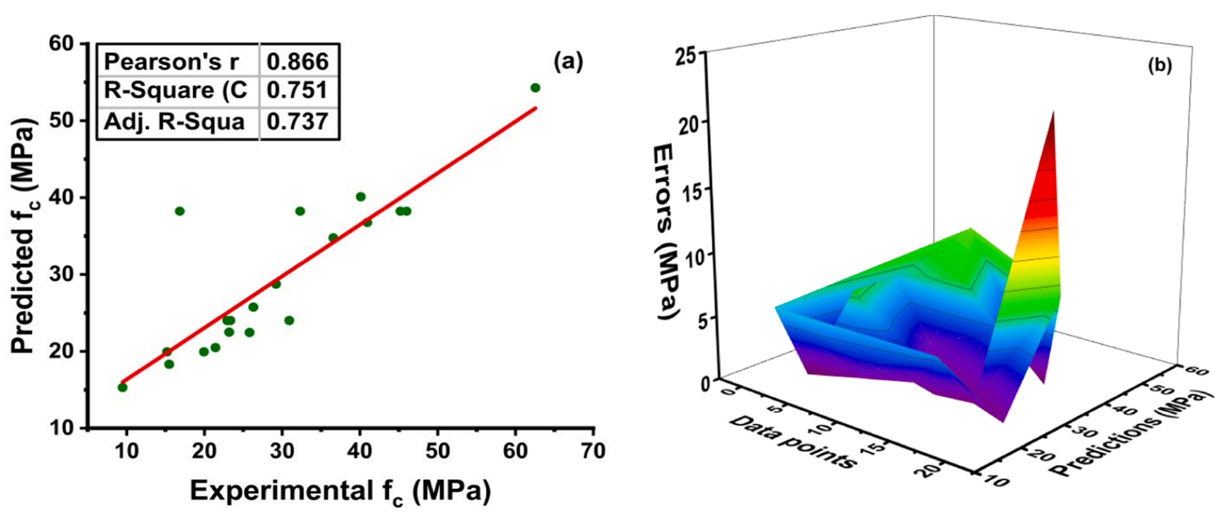

Fig. 8. Results obtained using DT algorithm: (a) Relation between the predicted and experimental values of the compressive strength; (b) Prediction errors performed by the DT algorithm.

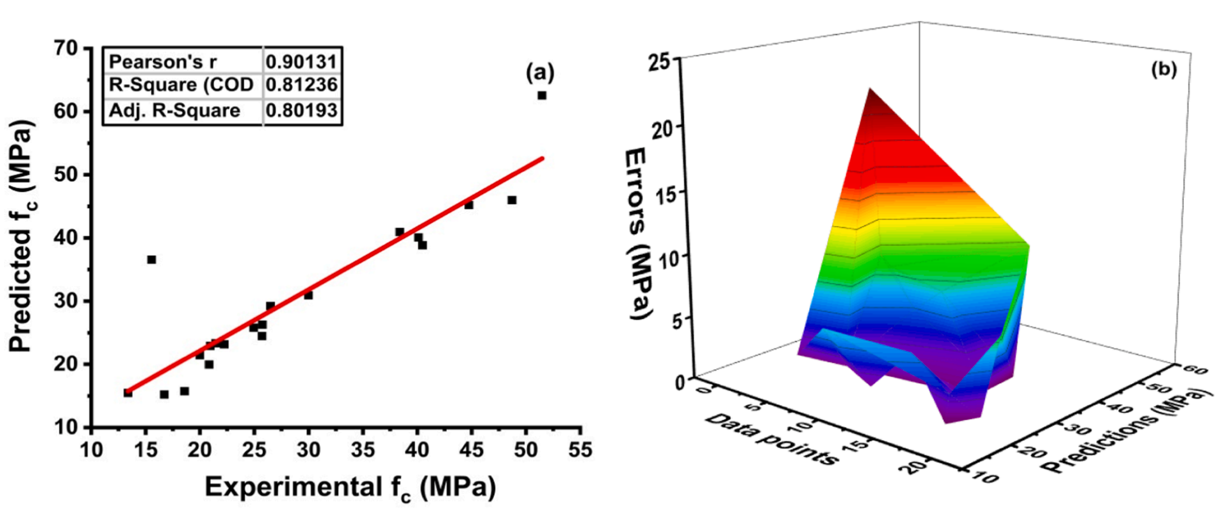

Fig. 9. Results obtained using ANN algorithm: (a) relation between the predicted and experimental values of the compressive strength; (b) Prediction errors performed by the ANN algorithm.

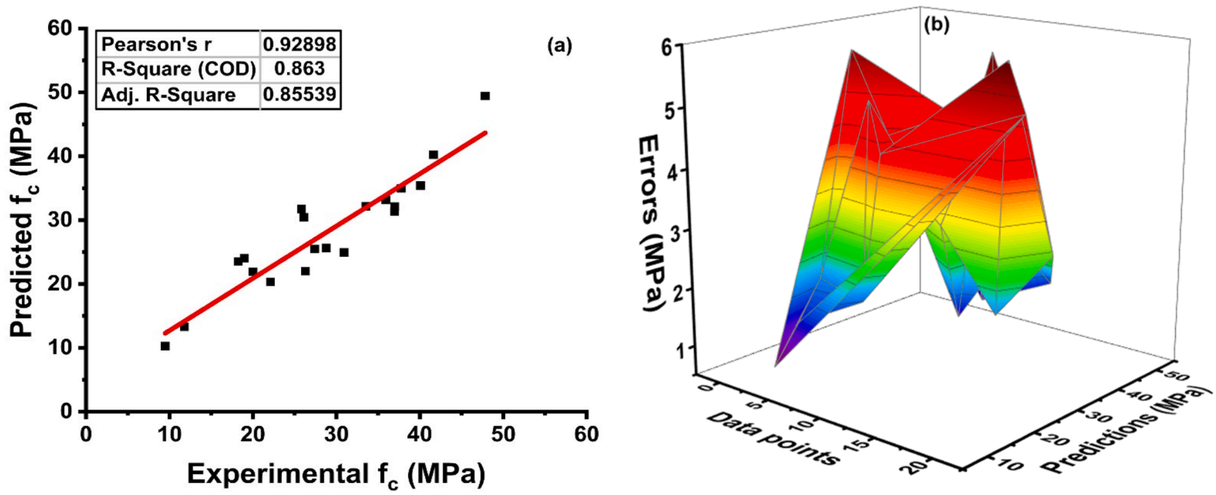

Fig. 10. Results obtained using GEP: (a) relation between the predicted and experimental values of the compressive strength; (b) Prediction errors performed by the GEP.

n = number of samples.

7.6. Statistical analysis and K fold analysis

A standard approach of K-fold cross validation is applied for examine the bias and variance decrease for the test set. To examine the result of cross-validation correlation coefficient (R^2), mean absolute error (MAE), mean square error (MSE), and Root mean square error (RMSE) were used as expressed in Fig. 11. In comparison, all three machine learning algorithms shows fluctuation in their output. The GEP model shows fewer errors with much better R^2 value oppose to ANN and Decision tree (DT). The average R^2 value of GEP modeling comes to 0.81 with the maximum of 0.96 and minimum values of 0.65, as illustrated in Fig. 11 (a). Whereas the ANN model gives the average value of $R^2 = 0.80$ of ten folds with the maximum and minimum values of 0.92 and 0.80, respectively. Similarly, the Decision tree (DT) model gives the average R^2 value of 0.78 with the maximum and minimum value of 0.90 and 0.58 as illustrate in Fig. 11(b) and (c). Every individual model reflects with the fewer errors for validation. As per the validation indicator result for the GEP reflect the values of MAE, MSE, and RMSE are 3.98 MPa, 25.57 MPa and 5.05 MPa reflected in Fig. 11(a). Indicator results for ANN gives the value of MAE, MSE, and RMSE are 4.48 MPa, 28.74 MPa and 5.36 MPa as illustrated in Fig. 11(b). Similarly, the values for decision tree of MAE, MSE and RMSE comes to 4.55 MPa, 32.66 MPa and 5.71 MPa respectively are shown in Fig. 11(c). While Fig. 11(d) illustrates the maximum value of R^2 for boosting regressor equals to 0.97, minimum 0.62 and average 0.83. The maximum values of MAE, MSE and, RMSE of BR were 8.98 MPa, 9.1 MPa and, 3.01 MPa, respectively. The minimum values (MAE, MSE and RMSE) for BR were 4.99 MPa, 5.2 MPa and, 2.28 MPa shown in Fig. 11(d). Moreover, the K-fold cross validation of all the applied models and statistical checks are also listed in the Table 9 and Table 10, respectively. Fig. 12 shows the statistical indicator for K-fold cross-validation. Table 11 indicates the values of statistical checks.

In addition, statistical checks applied on the models reveals that the ensemble ML algorithm shows lesser errors from other three employed individual algorithms (GEP, ANN, DT) as illustrated in the Table 10. The values of MAE, MSE, and RMSE of the bagging regressor (BR) from statistical checks comes to 3.69 MPa, 24.76 MPa, and 4.79 MPa, respectively. In comparison, the GEP, ANN, and DT shows sequence vice lesser error from one another. This check is directly linked with coefficient correlation (R^2), lesser the error value shows the higher R^2 value for the model.

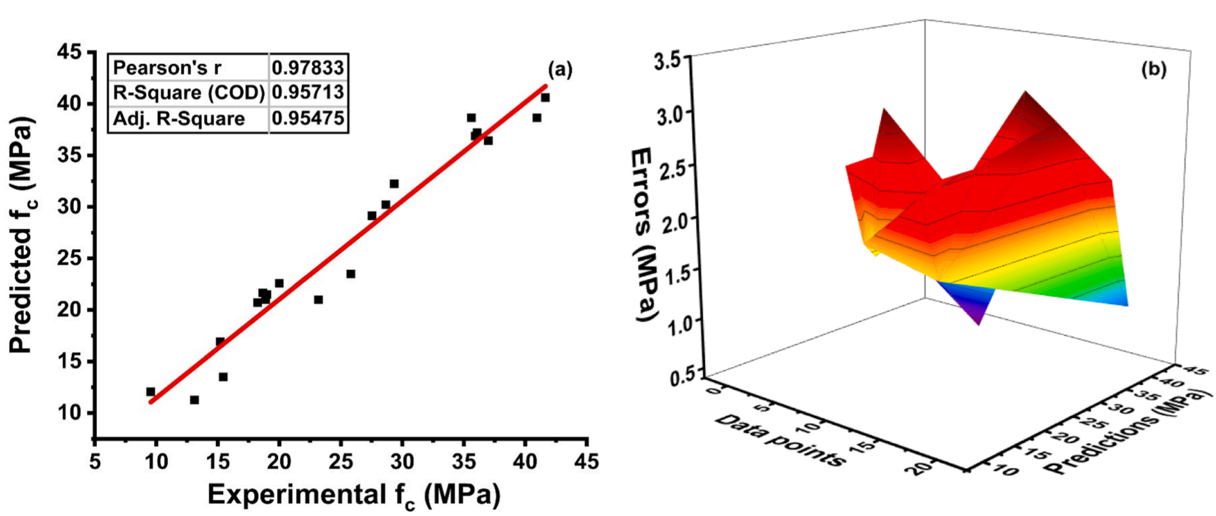

Fig. 11. Results obtained using Boosting Regressor: (a) represents the relationship between the predictied variable and the experimentally determined variable; (b) distribution of errors of boosting regressor.

Table 10
K-fold cross validation indicators and result.

	GEP				ANN				DT				Boosting			
K-fold	R^2	MAE	MSE	RMSE	R^2	MAE	MSE	RMSE	R2	MAE	MSE	RMSE	R2	MAE	MSE	RMSE
1	0.65	5.44	6.77	2.60	0.69	8.38	9.49	3.08	0.87	8.94	8.78	2.96	0.77	5.10	5.67	2.38
2	0.79	9.29	5.38	2.32	0.85	9.88	8.68	2.95	0.90	10.77	5.77	2.40	0.76	8.30	5.20	2.28
3	0.68	8.59	9.77	3.13	0.92	8.46	7.49	2.74	0.79	8.99	9.99	3.16	0.88	7.90	9.10	3.02
4	0.93	6.38	6.28	2.51	0.66	10.21	9.33	3.05	0.74	8.56	8.99	3.00	0.97	5.10	5.79	2.41
5	0.88	10.81	7.54	2.75	0.88	9.59	8.23	2.87	0.81	6.34	9.12	3.02	0.92	5.78	7.89	2.81
6	0.95	7.59	9.28	3.05	0.90	8.99	9.56	3.09	0.58	9.99	7.88	2.81	0.93	8.98	8.88	2.98
7	0.89	5.99	6.74	2.60	0.68	6.38	6.89	2.62	0.69	6.96	6.37	2.52	0.79	4.99	6.10	2.47
8	0.77	6.77	9.96	3.16	0.84	6.98	7.10	2.66	0.77	7.12	6.87	2.62	0.62	6.77	5.28	2.30
9	0.75	9.49	8.88	2.98	0.86	6.11	7.32	2.71	0.89	7.21	7.12	2.67	0.83	7.44	7.45	2.73
10	0.87	8.58	7.47	2.73	0.77	7.47	7.69	2.77	0.76	5.99	9.59	3.10	0.87	6.78	6.77	2.60

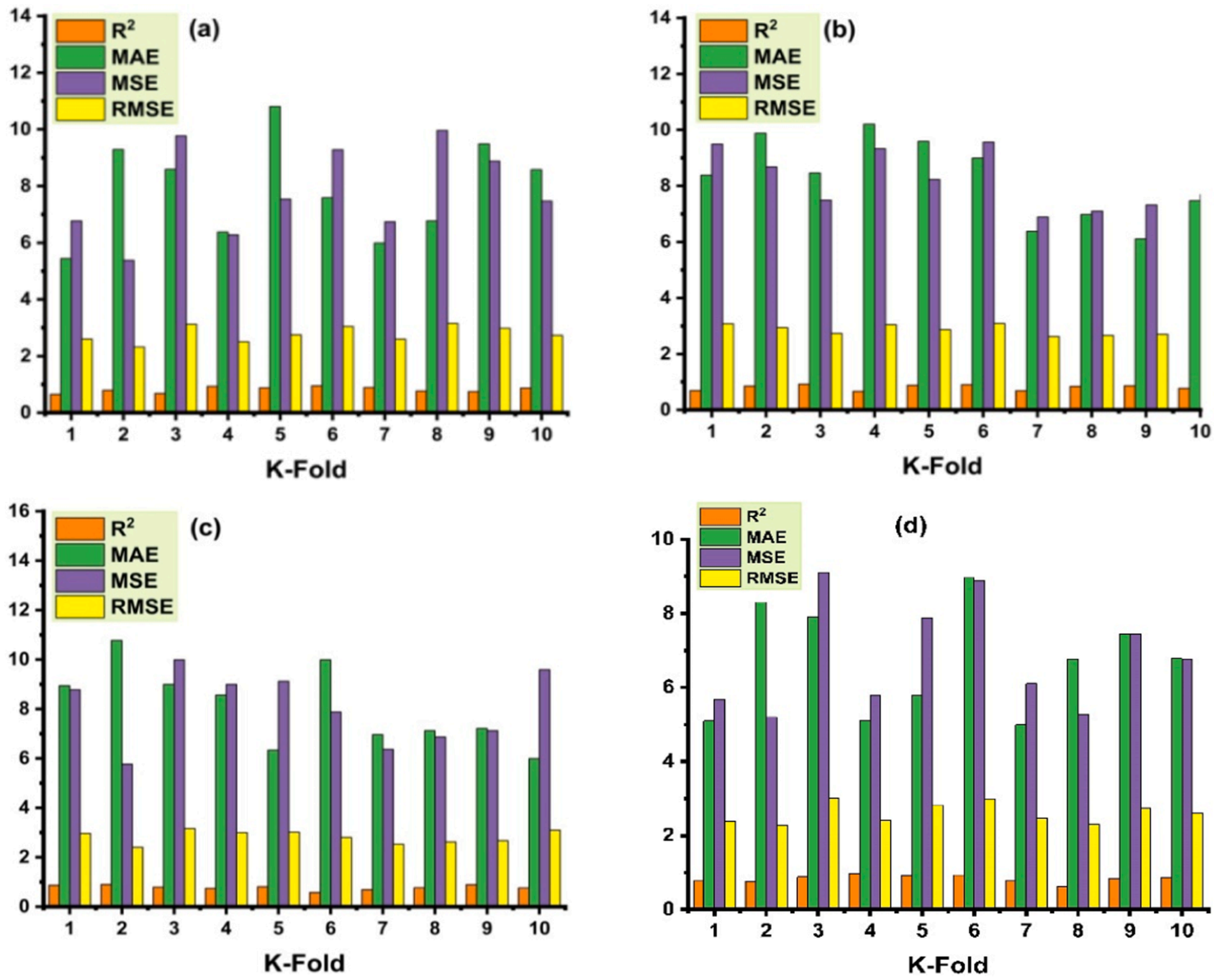

Fig. 12. Statistical indicator for K-fold cross-validation; (a) ANN model; (b) GEP model; (c) DT model; (d) Boosting regressor.

Table 11
Statistical Checks.

Machine Learning Techniques	MAE (MPa)	MSE (MPa)	RMSE (MPa)
Gene Expression Program (GEP)	3.98	25.57	5.05
Artificial Neural Network (ANN)	4.48	28.74	5.36
Decision Tree (DT)	4.55	32.66	5.71
Bagging Regressor (BR)	3.69	24.76	4.97

8. Conclusion

This research work is based on the performance comparison of the selected machine learning algorithms of concrete with fly ash as its main constituent. Supervised ML algorithms such as decision tree (DT), artificial neural network (ANN) and genetic engineering programming (GEP) and, bagging regressor (BR) were investigated for the prediction of compressive strength of fly ash-based concrete. Moreover, the individual ML algorithms were compared with ensemble ML approach for better understanding the performances of the approach.

1) The individual ML algorithms show better performance with less variance between the actual and the predicted outcomes.
2) While, when the accuracy level of the individual ML techniques was compared with ensemble employed algorithm (bagging regressor), the ensemble comes as a stronger and more accurate model indicated from its coefficient correlation (R^2) value equals to 0.95. The R^2

value of the GEP, ANN and, DT comes to 0.86, 0.81 and, 0.75, respectively.

3) The lesser values of the errors, MAE (3.69 MPa), MSE (24.76) and, RMSE (4.97) also confirmes the high accuracy of the bagging regressor, while other algorithms shows higher values for these errors.
4) K-fold cross validation process, which was adopted to confirm the model's accuracy also reveals that bagging regressor was effective.
5) Application of statistical checks also confirms that bagging regressor indicates the enhancement in model accuracy by minimizing the difference of error among the targeted and predicted results.

In conclusion, this study shows the reflection of various ML techniques for the predictive perspective. It gives the better understanding to the researcher in the field of engineering that the better selection of the input parameters and regressor to run the model would results in the accurate predicted outcomes. The ML approaches used in this work gives strong relation between the targeted and predicted results. Higher accuracy level of the models indicates the importance of their use in the field of civil engineering, especially when it comes to predict the strength properties of concrete, because practically it is time consuming task to have a strength result of the concrete. This approach is now grooming the scientific society by minimizing the time and physical activities. Moreover, the application of more ensemble algorithms like, Adaboost regressor, XGboost regressor and bagging regressor would be more effective for the log list of the datapoints.

CRediT authorship contribution statement

Hongwei Song: Conceptualization, Methodology, Investigation, Funding acquisition. **Ayaz Ahmad:** Formal analysis, Software, Investigation, Visualization, Writing – original draft, Writing – review & editing. **Furqan Farooq:** Formal analysis, Software, Writing – original draft. **Krzysztof Adam Ostrowski:** Methodology, Writing – original draft, Writing – review & editing. **Mariusz Maślak:** Conceptualization, Writing – original draft, Writing – review & editing. **Slawomir Czarnecki:** Methodology, Writing – original draft, Writing – review & editing. **Fahid Aslam:** Formal analysis, Software, Visualization, Writing – original draft.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Annexure a

Cement (kg/m^3)	Fly ash (kg/m^3)	Water (kg/m^3)	Superplasticizer (kg/m^3)	Coarse aggregate (kg/m^3)	Fine aggregate (kg/m^3)	Age (Days)	Strength (MPa)
185.7	101.5	166.9	7.5	1006.4	905.4	90	38.28
170.7	127.4	161.8	7.8	1090	798.5	3	17.11
180.7	127.4	166.1	7.8	1090	798.6	14	23.34
160.7	127.4	162.1	7.8	1090	804	28	27.41
242.1	125.6	184.3	5.7	1057.6	779.3	14	21.91
212.1	125.6	184.2	5.7	1057.6	779.3	28	21.81
212.1	124.6	184.2	5.7	1057.6	779.3	56	33.21
240	118.3	191.9	4.6	1029.4	758.6	90	32.31
195.3	124.2	160.8	9.9	1088.1	802.6	3	9.49
190.7	124.2	161.9	9.9	1088.1	802.6	14	23.1
168.1	168.3	172.4	4.5	1058.6	780.1	3	11.78
136.1	163.3	176.5	4.5	1058.6	780.1	14	27.49
165.2	161.3	172.4	4.5	1058.6	780.1	28	25.2
230.6	118.2	195.2	6.1	1028.1	757.6	3	15.47
228.7	119.2	191.3	6.1	1028.1	757.6	14	21.31
229.7	118.2	195.2	6.1	1028.1	757.6	90	42.37
248.1	94.1	186.7	7	949.9	847	3	21.43
238.1	92.1	188.7	7	949.9	847	14	26.6
248.1	92.1	186.7	7	949.9	847	90	47.83
250.5	95.7	187.4	5.5	956.9	861.2	3	12.78
251	95.7	186.5	5.5	956.9	861.2	14	27.4
212.5	124.9	159	7.8	1085.4	799.5	3	18.42
212.6	124.7	181.9	5.8	1028.4	757.7	14	30.92
251.8	124.2	188.5	6.4	1028.4	810.7	56	34.9
251.9	122.8	181.6	6.4	1028.4	757.7	90	45.2
181.8	122.8	169.6	7.6	1055.6	810.7	3	14.61
181.8	122.8	170.6	7.6	1055.6	810.7	14	22.93
181.8	122.8	169.6	7.6	1055.6	777.8	28	28.79
182.5	124.4	170.6	7.6	1055.6	777.8	56	37.77
181.4	121.8	169.6	7.6	1055.6	777.8	90	46.77
250.1	124.8	168.3	9.4	961.2	865	90	45.99
230.1	124.8	160.6	11.8	973.9	875.6	3	22.11
220.8	124.8	146.1	12.4	1006	899.8	28	31.74
210.8	124.8	143.3	12	1086.8	800.9	56	62.55
220.8	124.7	141.1	12	1086.8	800.9	90	62.32
213.7	124.7	154.8	10.2	1053.5	776.4	28	43.15
213.7	124.8	155	10.2	1053.5	776.4	56	47.14
213.7	124.8	154.8	10.2	1053.5	776.4	90	54.09
213.5	124.8	155.6	11.7	1052.3	775.5	3	19.64
213.5	124.8	154.6	11.7	1052.3	800	14	36.98
213.5	124.8	155.7	11.7	1052.3	800	28	42.44

(continued on next page)

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Cement (kg/m^3)	Fly ash (kg/m^3)	Water (kg/m^3)	Superplasticizer (kg/m^3)	Coarse aggregate (kg/m^3)	Fine aggregate (kg/m^3)	Age (Days)	Strength (MPa)
213.5	124.8	154.6	11.7	1052.3	800.5	56	49.43
213.5	124.8	155.5	11.7	1052.3	800	90	58.4
218.9	124.8	158.5	11.3	1078.7	794.9	56	40.21
218.9	124.8	160.1	11.3	1078.7	794.9	90	45.25
376	124.8	216.7	0	1003.5	762.4	3	18.98
190.3	124.8	165.6	9.9	1079	798.9	14	20.22
165	124.8	163.8	0	1005.6	900.9	28	26.11
190.3	124.8	165.3	9.9	1079	798.9	28	26.67
250	124.8	192.8	5.3	948.9	857.2	28	28.21
213.5	124.8	159.2	11.7	1043.6	771.9	28	44.61
194.7	124.8	171.5	7.5	998	901.8	28	39.27
251.4	124.8	192.9	5.8	1043.6	754.3	28	36.99
310	124.8	189.9	0	936.2	712.2	28	40.94
280	124.8	189.9	0	936.2	700	7	36.57
290.2	124.8	183.6	0	1069.2	754.3	7	23.18
252.5	124.8	186.7	0	1111.6	784.3	7	13.11
339	124.8	197	0	968	800	3	18.24
257	124.8	192.8	0	968	856.5	90	27.54
254	124.8	192.7	0	968	799.7	90	28.66
307	124.8	193.5	0	968	799.5	28	29.33
307	124.8	191.2	0	968	799.5	90	35.92
290	124.8	192.2	0	936	755	28	46.29
297	124.8	191.3	0	936	755	90	51.18
299	124.8	187.3	0	966	763	3	17.11
288	124.8	188.6	0	966	758	7	21.21
289	124.8	188.6	0	966	759	14	29.22
292	124.8	187.3	0	966	763	28	33.55
331	124.8	192.2	0	978	801	90	40.1
349	124.8	192.2	0	1047	806	3	16.59
295	124.8	185.3	0	1069	769.4	28	27.19
238	124.8	185.2	0	1118	789	28	16.84
296	124.7	191.3	0	1085	765.5	7	16.83
322.5	124.8	203.4	0	974	840	14	24.11
322	124.4	201.5	0	974	800.2	28	26.15
322	124.7	202.7	0	974	820	90	30.57
302	124.8	202.7	0	974	817	28	26.11
312.5	124.6	182.4	0	1040	734	28	40.08
317	124.8	192.5	0	936	721	3	26.29
210	124.8	142.5	0	896	896	7	49.41
220.9	124.8	142.5	0	896	896	28	72.11
144	124.8	158.2	18	943	844	28	17.42
148	124.8	158.4	16	1002	830	28	19.95
326	124.6	199.1	11	801	792	28	39.78
290	124.8	220.5	11	898	713	28	9.59
299.8	124.8	211.5	9.9	878.2	727.6	28	25.81
148.1	124.6	159.2	16.1	1001.8	830.1	28	18.98
326.5	124.8	193.3	10.8	801.1	792.5	28	35.61
276.4	124.8	217.4	11	897.7	712.9	28	9.55
150.7	124.8	164.6	15.6	1074.5	687	28	15.44
190.8	124.8	185.1	11.1	979.5	811	28	15.21
190.9	124.8	168.2	11.6	991.2	784	28	18.66
188.7	124.8	182.4	11.7	1023.3	732	28	20.01
298.1	124.6	189.4	6.1	879	815	28	41.64
318.9	124.8	212.7	5.7	860.5	725	28	36.09
355.9	124.5	196.3	11	801.4	769	28	36.27
199.8	124.8	185.4	12.6	849.3	856.5	28	18.01
278.7	124.8	170.3	10.1	925.3	782	28	41.16

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